WORKED SOLUTION

STEP III 2020, Question 3

Pure •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

In the complex plane a rotation about a through angle θ is just $z \mapsto a + (z-a)e^{i\theta}$, so G_{AB} is an explicit linear combination of a and b. Once part (i) gives that formula, the geometry turns into algebra: a parallelogram is a linear condition on the vertices, and an equilateral triangle is one side being another side turned through a fixed angle. The whole question rests on a few facts about powers of ω.

Solution

Throughout, $\omega = e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$, so $\omega^* = e^{-i\pi/6}$, and we write

$$\rho = e^{-i\pi/3} = \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i.$$

Three facts we shall use:

$$\omega^2 = e^{i\pi/3}, \qquad \omega^3 = i, \qquad \omega - \omega^* = 2i\sin\tfrac{\pi}{6} = i.$$

Part (i)

A clockwise rotation through $\frac{1}{3}\pi$ multiplies by ρ. Rotating about A means we rotate the displacement $b - a$ and add it back on to a, so

$$k = a + (b-a)\rho.$$

The centroid of ABK is therefore

$$g_{ab} = \tfrac{1}{3}(a+b+k) = \tfrac{1}{3}[(2-\rho)\,a + (1+\rho)\,b].$$

Now we tidy the two coefficients:

$$2 - \rho = \tfrac{3}{2} + \tfrac{\sqrt{3}}{2}i = \sqrt{3}\left(\tfrac{\sqrt{3}}{2} + \tfrac{1}{2}i\right) = \sqrt{3}\,\omega,$$
$$1 + \rho = \tfrac{3}{2} - \tfrac{\sqrt{3}}{2}i = \sqrt{3}\left(\tfrac{\sqrt{3}}{2} - \tfrac{1}{2}i\right) = \sqrt{3}\,\omega^*.$$

Hence

$$g_{ab} = \tfrac{1}{3}(\sqrt{3}\,\omega a + \sqrt{3}\,\omega^* b) = \frac{1}{\sqrt{3}}(\omega a + \omega^* b),$$

as required.

Part (ii)

A quadrilateral $PQRS$ (vertices in that order) is a parallelogram exactly when $\overrightarrow{PQ} = \overrightarrow{SR}$, that is, $q - p = r - s$, or equivalently

$$p - q + r - s = 0.$$

(This is the same as saying the diagonals PR and QS have the same midpoint.) So Q_1 is a parallelogram if and only if $a - b + c - d = 0$, and Q_2 is a parallelogram if and only if

$$g_{ab} - g_{bc} + g_{cd} - g_{da} = 0.$$

Using part (i) for each of the four points,

$$\begin{aligned}\sqrt{3}\,(g_{ab} - g_{bc} + g_{cd} - g_{da}) &= (\omega a + \omega^* b) - (\omega b + \omega^* c) + (\omega c + \omega^* d) - (\omega d + \omega^* a)\\ &= (\omega - \omega^*)a - (\omega - \omega^*)b + (\omega - \omega^*)c - (\omega - \omega^*)d\\ &= \mathrm{i}\,(a - b + c - d).\end{aligned}$$

Thus

$$g_{ab} - g_{bc} + g_{cd} - g_{da} = \frac{\mathrm{i}}{\sqrt{3}}\,(a - b + c - d),$$

and since $\mathrm{i}/\sqrt{3} \neq 0$, one side vanishes if and only if the other does. Therefore Q_1 is a parallelogram if and only if Q_2 is.

Ask yourself Could Q_2 satisfy the parallelogram condition but collapse, with two of its vertices equal?

Yes, in one special case. If ABC is an equilateral triangle labelled clockwise, then rotating B clockwise about A lands on C, so G_{AB} and G_{BC} are both the centroid of ABC. Completing the rhombus $ABCD$, Q_2 flattens to a segment. The condition $p - q + r - s = 0$ still holds there, so the equivalence is really a statement about this vector condition, which is what "parallelogram" means in a question like this.

Part (iii)

We show that one side of T_2, as a vector, is another side turned through a fixed angle. From part (i),

$$\begin{aligned}\sqrt{3}\,(g_{bc} - g_{ab}) &= (\omega b + \omega^* c) - (\omega a + \omega^* b) = -\omega a + \mathrm{i}\,b + \omega^* c,\\ \sqrt{3}\,(g_{ca} - g_{bc}) &= (\omega c + \omega^* a) - (\omega b + \omega^* c) = \omega^* a - \omega b + \mathrm{i}\,c,\end{aligned}$$

using $\omega - \omega^* = \mathrm{i}$ in each line.

Ask yourself What number λ would turn the second bracket into the first?

Matching the coefficients of a forces $\lambda\omega^* = -\omega$, so $\lambda = -\omega/\omega^* = -\omega^2 = -\mathrm{e}^{\mathrm{i}\pi/3} = \mathrm{e}^{-2\pi\mathrm{i}/3}$. This same λ works for b and c too:

$$\lambda \cdot (-\omega) = \omega^3 = \mathrm{i},$$
$$\lambda \cdot \mathrm{i} = -\omega^2 \cdot \omega^3 = -\omega^5 = -\mathrm{e}^{5\pi\mathrm{i}/6} = \mathrm{e}^{-\mathrm{i}\pi/6} = \omega^*.$$

Hence, for all a, b, c,

$$g_{bc} - g_{ab} = \mathrm{e}^{-2\pi\mathrm{i}/3}\,(g_{ca} - g_{bc}).$$

Taking moduli, $G_{AB}G_{BC} = G_{BC}G_{CA}$, since $|\mathrm{e}^{-2\pi\mathrm{i}/3}| = 1$. Also the direction of travel along T_2 turns through $\frac{2}{3}\pi$ at G_{BC}, so the interior angle of T_2 at G_{BC} is $\pi - \frac{2}{3}\pi = \frac{1}{3}\pi$. A triangle with two equal sides enclosing an angle of $\frac{1}{3}\pi$ is isosceles with base angles $\frac{1}{2}(\pi - \frac{1}{3}\pi) = \frac{1}{3}\pi$, so all its angles are $\frac{1}{3}\pi$ and T_2 is equilateral.

The one exception is when $g_{ca} = g_{bc}$, and then the relation gives $g_{ab} = g_{bc}$ as well, so T_2 shrinks to a single point. Multiplying $\omega^* a - \omega b + \mathrm{i}c = 0$ by ω gives $a - \mathrm{e}^{\mathrm{i}\pi/3}b + \mathrm{e}^{2\pi\mathrm{i}/3}c = 0$, which happens exactly when ABC is itself an equilateral triangle labelled clockwise (as in the remark in part (ii)). Every other triangle T_1, including degenerate ones, gives a genuine equilateral T_2.

Checks

- Take $a = 0$, $b = 1$. Then $k = \rho$, and the centroid of $0, 1, \rho$ is $\frac{1}{3}(1 + \rho) = \frac{1}{2} - \frac{\sqrt{3}}{6}\mathrm{i}$. The formula gives $\omega^*/\sqrt{3} = \frac{1}{2} - \frac{1}{2\sqrt{3}}\mathrm{i}$, the same number.
- With $a = 0$, $b = 1$, $c = \mathrm{i}$ (a right-angled triangle), the three points of T_2 have pairwise distances all equal to about 1.1154. A computer check on 10^4 random triangles and quadrilaterals confirmed the identities in parts (ii) and (iii) to rounding error.

Common slips

- Rotating b about the origin, writing $k = b\rho$, rather than about A.
- Using $\mathrm{e}^{+\mathrm{i}\pi/3}$ for a clockwise rotation. Clockwise is the negative sense.
- In part (ii), proving only one direction. The factor $\mathrm{i}/\sqrt{3}$ being non-zero is what gives "if and only if".

Beyond STEP

Part (iii) is Napoleon's theorem. Erect equilateral triangles on the sides of any triangle (here outwards, when ABC is labelled anticlockwise), and their centroids form an equilateral triangle. The complex-number proof works because rotations and centroids are both linear maps of the plane. That is the starting point for the study of transformation

groups in geometry. Needham's *Visual Complex Analysis* treats this geometric view of complex multiplication beautifully.

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